Exercise 4.1

(1)(i) 17 (ii) 6 (iii) 20 (iv) 720 (v) 120

(2)(i) 151200 (ii) 24 (3)(i) 12 (ii) 24 (4)(i) 64 (ii) 24

(5)(i) 90 (ii) 64 (6)(i) 144 (ii) 80 (7)(i) 48 (ii) 90

(8)(i) 9000 (ii) 4536 (iii) 4464 (9)(i) 36 (ii) 60 (10) 13

(11) 400 (12)(i) 42 (ii) 78 (13)(i) 4^6 (ii) 3^{10} (iii) 10^{12}

(14)(i) 720 (ii) 144 (iii) 4 (iv) 144 (v) 220 (vi) $(n+3)(n+2)$

(15)(i) 15 (ii) 120 (iii) $\frac{n(n-1)}{2}$ 16.(i) 4 (ii) 100

Exercise 4.2

(1) 10 (2) 4 (3)(i) 336 (ii) 172800 (4) 720

(5)(i) 4^{10} (ii) $3^4 \times 5^6$ (iii) 11! (6)(i) 4^5 (ii) 15^5

(7) 144 (8)(i) 14! (ii) $9! \times 6!$ (iii) $8! \times {}^9P_6$ (9) 34650

(10) 1260 (11) 6912 (12) 60 (13)(i) 2^8 (ii) 28

(14)(i) 43200 (ii) 151200 (iii) 19807200 (iv) 151200 (15)(i) 180

(ii) 60 (iii) 30 (16)(i) 379 (ii) 135 (17) 120, NIGHT

(18) 7 (19) 399960 (20) 571956

Exercise 4.3

(1) 1 (2) 3 (3) 10, 3 (6) 20

(9) (i) ${}^{14}C_7 = 3432$ (ii) ${}^{15}C_2 = 105$ (iii) ${}^{20}C_2 = 190$ (iv) ${}^{100}C_5$ (v) ${}^5C_3 \times {}^4C_2 \times {}^2C_1 = 120$

(10) (i) $2^4 = 16$ (ii) $2^5 = 32$ (iii) 2^n

(11) (i) ${}^{25}C_3$ (ii) ${}^{25}P_3$

(12) ${}^{10}P_2 \times {}^8C_4 = 6300$ (13)(i) ${}^{10}C_3 = 120$ (ii) ${}^{10}C_5 = 252$

(14) ${}^5C_2 \times {}^{20}C_3 = 11400$ (i) ${}^4C_1 \times {}^{20}C_3 = 4560$ (ii) ${}^5C_2 \times {}^{19}C_3 = 9690$

(15) ${}^7C_3 = 35$ (16) 546 (17)(i) 280 (ii) 336 (iii) 736

(18) 485 (19) 64 (20) 2454

(21) ${}^{15}C_3 = 455$ (22) 364 (23)(i) 50 (ii) 161 (24) 15